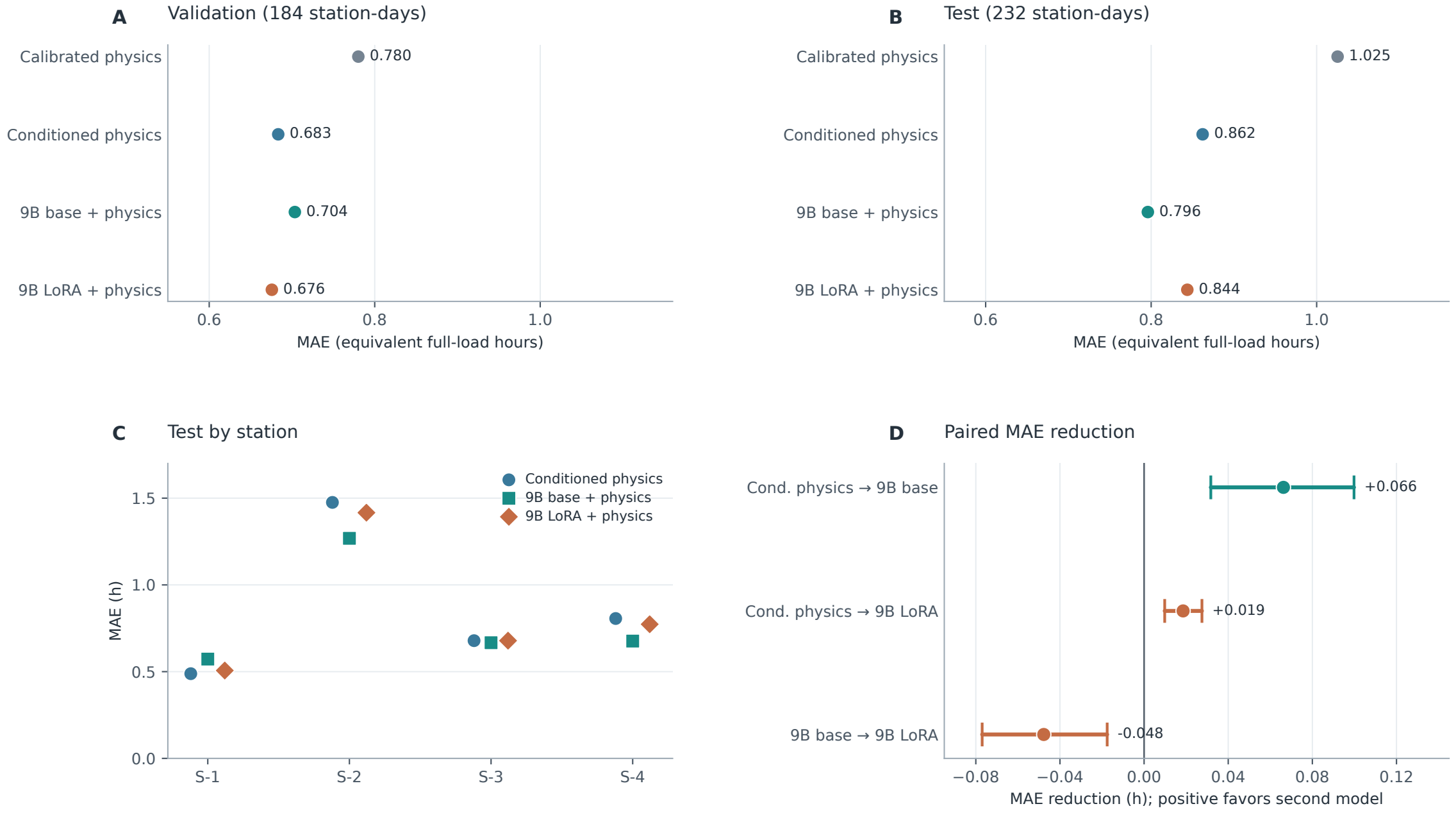
A
Validation (184 station-days)
Calibrated physics
Conditioned physics
9B base + physics
9B LoRA + physics
0.780
0.683
0.704
0.676
0.6
0.8
1.0
MAE (equivalent full-load hours)
B
Test (232 station-days)
Calibrated physics
Conditioned physics
9B base + physics
9B LoRA + physics
1.025
0.862
0.796
0.844
0.6
0.8
1.0
MAE (equivalent full-load hours)
C
Test by station
Conditioned physics
9B base + physics
9B LoRA + physics
MAE (h)
0.0
0.5
1.0
1.5
S-1
S-2
S-3
S-4
D
Paired MAE reduction
Cond. physics → 9B base
Cond. physics → 9B LoRA
9B base → 9B LoRA
+0.066
+0.019
-0.048
−0.08
−0.04
0.00
0.04
0.08
0.12
MAE reduction (h); positive favors second model